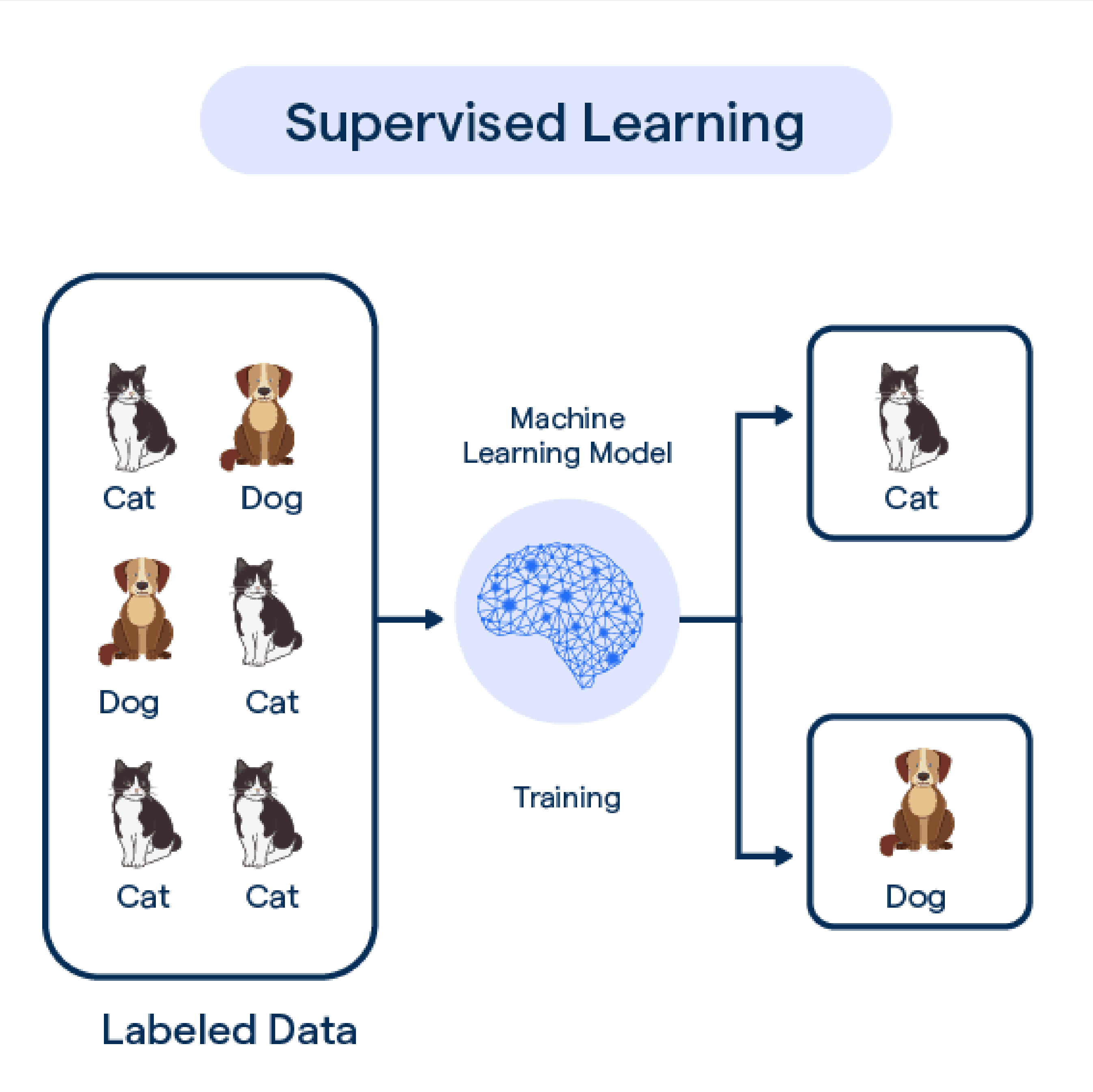
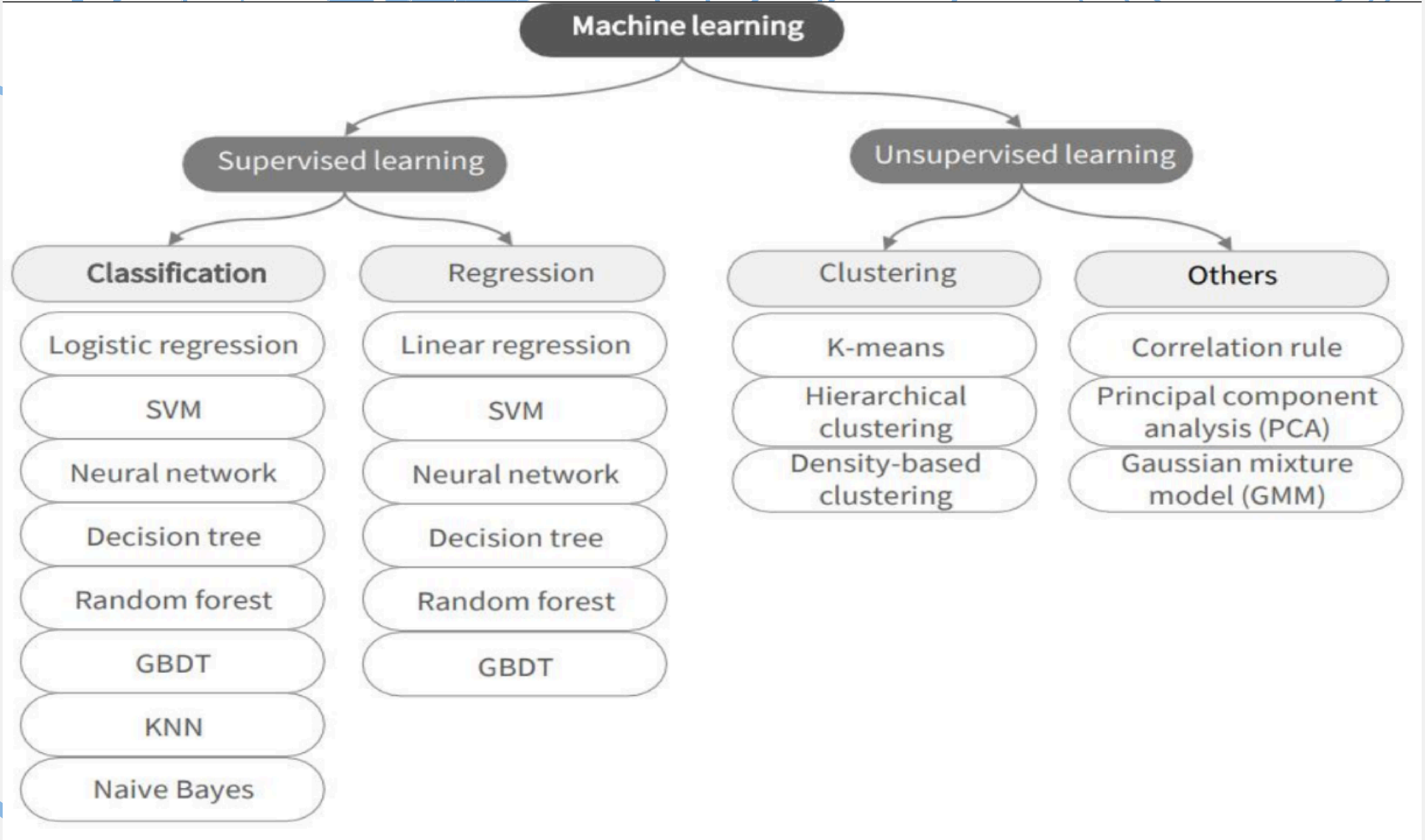
Supervised Learning

MSP’26 ML Module Session02

Supervised Learning

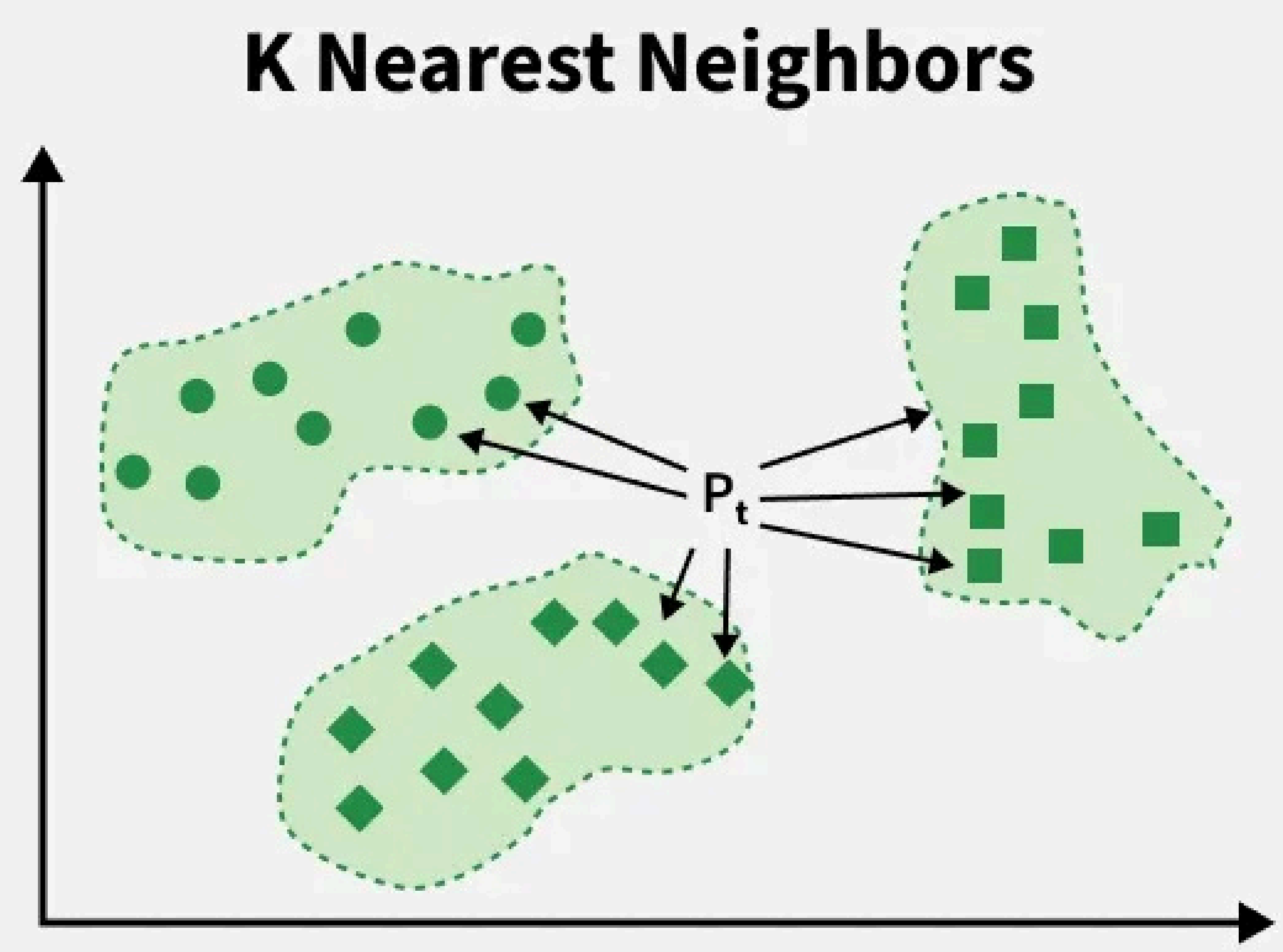
Supervised learning is a type of machine learning where a model learns from labelled data—meaning every input has a corresponding correct output. The model makes predictions and compares them with the true outputs, adjusting itself to reduce errors and improve accuracy over time. The goal is to make accurate predictions on new, unseen data.




Machine learning
Supervised learning
Unsupervised learning
Classification
Logistic regression
SVM
Neural network
Decision tree
Random forest
GBDT
KNN
Naive Bayes
Regression
Linear regression
SVM
Neural network
Decision tree
Random forest
GBDT
Clustering
K-means
Hierarchical clustering
Density-based clustering
Others
Correlation rule
Principal component analysis (PCA)
Gaussian mixture model (GMM)

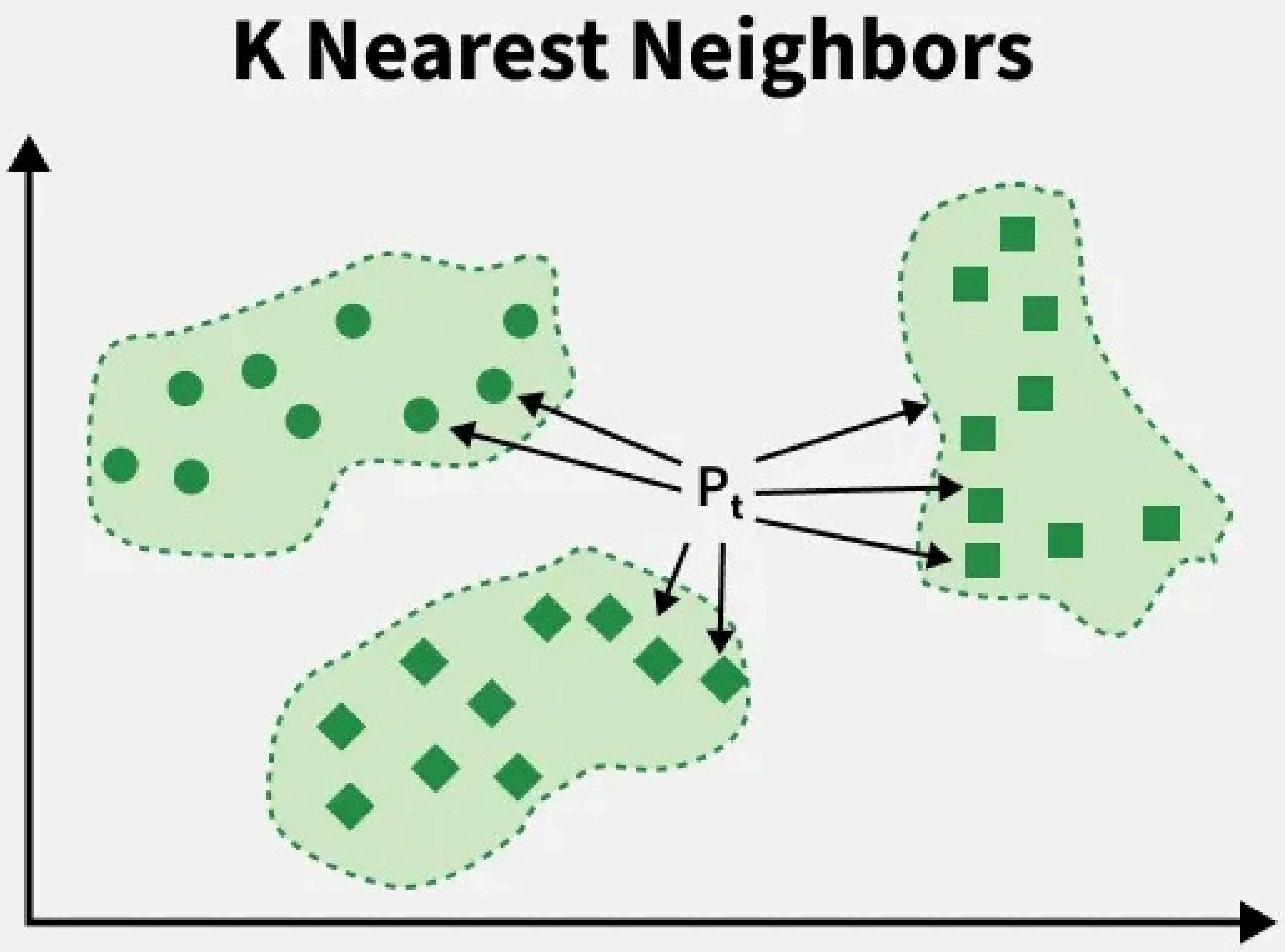
K-nearest neighbours (knn)

1. Initialize k to your chosen number of neighbours
2. For each sample in the data:
- Calculate the distance between the query example and the current example from the data
- Add the distance and the index of the example to an ordered collection
3. Sort the ordered collection of distances and indices from smallest to largest



K-nearest neighbours (knn) cont'd

4.Sort the ordered collection of distances and indices from smallest to largest (in ascending order) by the distances
5.Pick the first K entries from the sorted collection
6.Get the labels of the selected K entries
7.if regression, return the mean of the K labels
8. If classification, return the mode of the K labels (if using majority voting)



knn – classification

Training Point (X1, X2)	Class	Distance from (3,7)	Calculation	Rank
(7, 7)	Bad	4.00	$\sqrt{[(7-3)^2 + (7-7)^2]} = \sqrt{(16+0)} = 4.00$	3
(7, 4)	Bad	5.00	$\sqrt{[(7-3)^2 + (4-7)^2]} = \sqrt{(16+9)} = 5.00$	4
(3, 4)	Good	3.00	$\sqrt{[(3-3)^2 + (4-7)^2]} = \sqrt{(0+9)} = 3.00$	1
(1, 4)	Good	3.61	$\sqrt{[(1-3)^2 + (4-7)^2]} = \sqrt{(4+9)} = 3.61$	2

knn – classification cont'd

Step 3: Select K = 3 nearest neighbors

Neighbor	Class
1st nearest: (3,4)	Good
2nd nearest: (1,4)	Good
3rd nearest: (7,7)	Bad

Step 4: Majority voting

Class	Count
Good	2
Bad	1

knn – regression

ID	Height	Age	Weight	Distance from (5.5,38)	Calculation	Rank
1	5.0	45	77	7.02	$\sqrt{[(5.0-5.5)^2+(45-38)^2]} = \sqrt{(0.25+49)}$	9
2	5.11	26	47	12.00	$\sqrt{[(5.11-5.5)^2+(26-38)^2]} = \sqrt{(0.15+144)}$	10
3	5.6	30	55	8.00	$\sqrt{[(5.6-5.5)^2+(30-38)^2]} = \sqrt{(0.01+64)}$	7
4	5.5	34	59	4.00	$\sqrt{[(5.5-5.5)^2+(34-38)^2]} = \sqrt{(0+16)}$	2
5	4.8	40	72	2.12	$\sqrt{[(4.8-5.5)^2+(40-38)^2]} = \sqrt{(0.49+4)}$	1
6	5.8	36	60	2.02	$\sqrt{[(5.8-5.5)^2+(36-38)^2]} = \sqrt{(0.09+4)}$	0
7	5.3	19	40	19.00	$\sqrt{[(5.3-5.5)^2+(19-38)^2]} = \sqrt{(0.04+361)}$	11
8	5.8	28	60	10.00	$\sqrt{[(5.8-5.5)^2+(28-38)^2]} = \sqrt{(0.09+100)}$	8
9	5.5	23	45	15.00	$\sqrt{[(5.5-5.5)^2+(23-38)^2]} = \sqrt{(0+225)}$	12
10	5.6	32	58	6.00	$\sqrt{[(5.6-5.5)^2+(32-38)^2]} = \sqrt{(0.01+36)}$	6

knn – regression cont'd

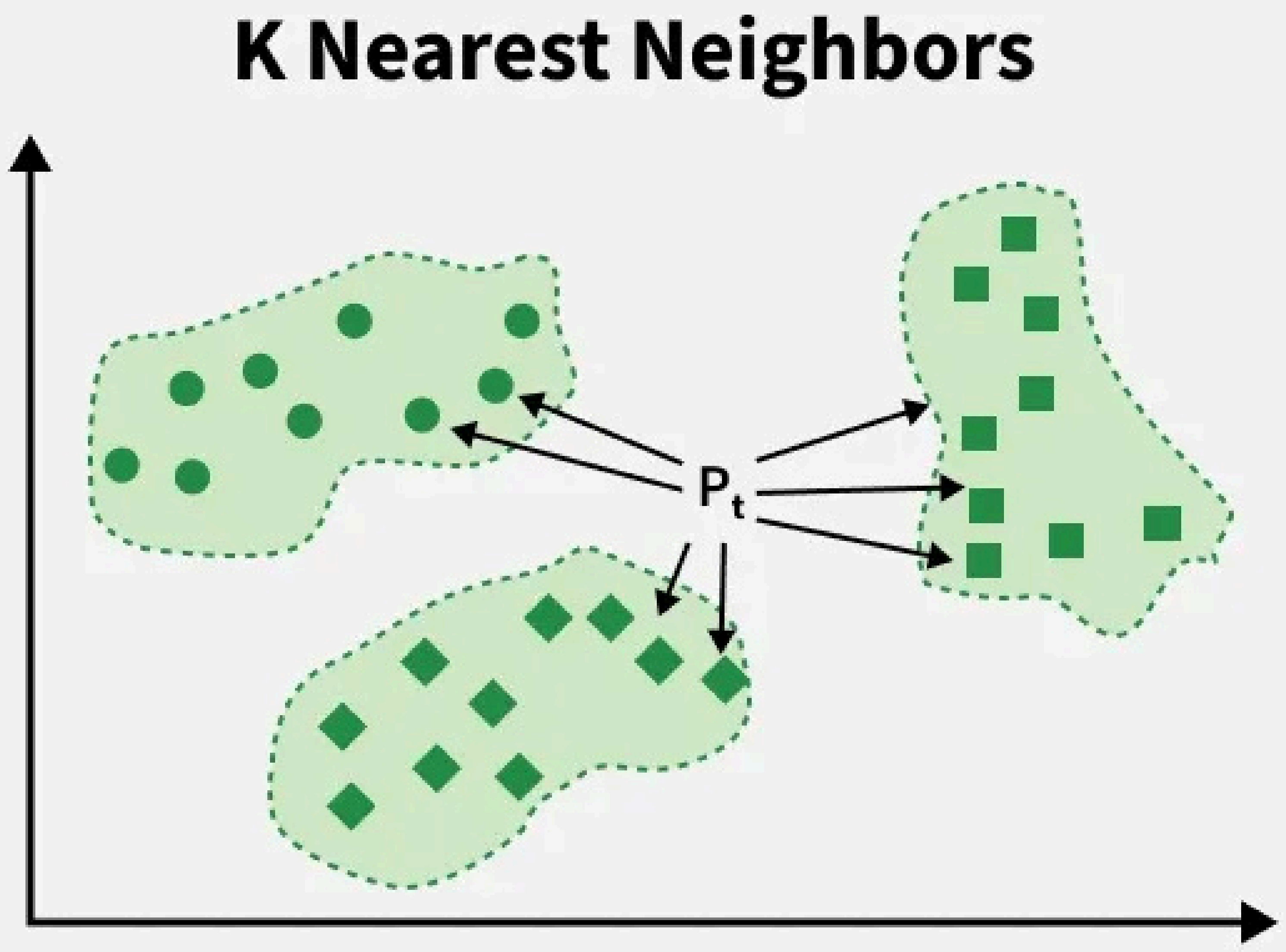
Nearest 3 distances:

ID	Weight	Distance
6	60	2.02
5	72	2.12
4	59	4.00

$$Predicted\ Weight = \frac{60 + 72 + 59}{3}$$
$$= \frac{191}{3}$$
$$= 63.67$$

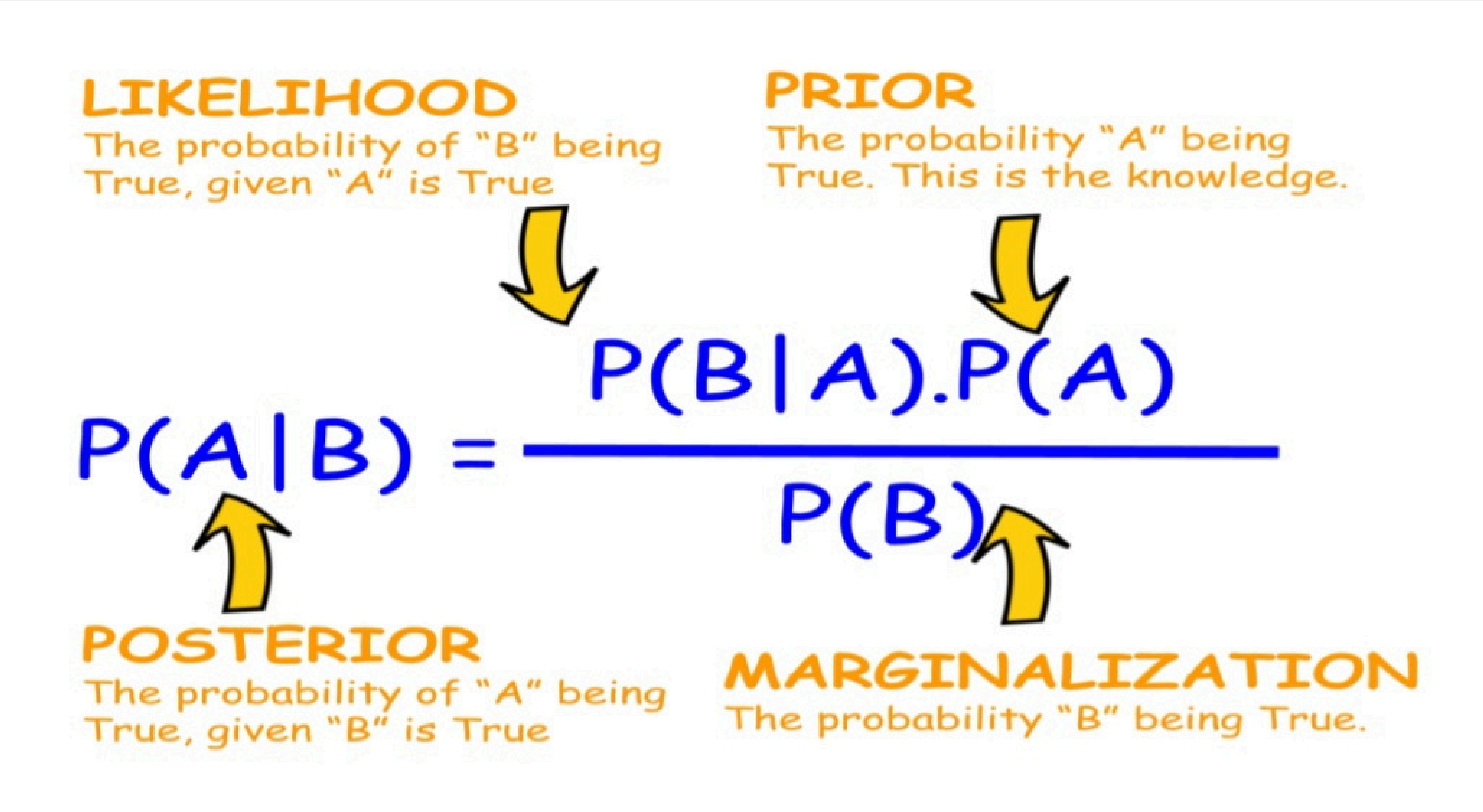
knn – how to pick optimal k

- **Small K values:** This can make the model highly sensitive to noise and might overfit the training data
- **Large K values:** This can make the model smooth out the decision boundary, leading to underfitting. The classifier becomes less sensitive to local patterns and more influenced by the overall data distribution.



Search Task: how to pick perfect k?

Naive Bayes Classifier



Naive Bayes – categorical

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

$\textit{New Instance} = (\textit{Red}, \textit{SUV}, \textit{Domestic})$

Naive Bayes – categorical cont'd

$$p(Yes) = \frac{5}{10} = 0.5$$

$$p(No) = \frac{5}{10} = 0.5$$

Color	Yes	No
Red	3/5	2/5
Yellow	2/5	3/5

Type	Yes	No
Sports	4/5	2/5
SUV	1/5	3/5

Origin	Yes	No
Domestic	2/5	3/5
Imported	3/5	2/5

$$P(Yes|New\ Instance) = p(Yes) * P(Color = Red|Yes) * P(Type = SUV|Yes) * P(Origin = Domestic|Yes)$$

$$P(Yes|New\ Instance) = \frac{5}{10} * \frac{3}{5} * \frac{1}{5} * \frac{2}{5} = \frac{3}{125} = 0.024$$

$$P(No|New\ Instance) = p(No) * P(Color = Red|No) * P(Type = SUV|No) * P(Origin = Domestic|No)$$

$$P(No|New\ Instance) = \frac{5}{10} * \frac{2}{5} * \frac{3}{5} * \frac{3}{5} = \frac{9}{125} = 0.072$$

Naive Bayes – numerical

Class	Temperature
Healthy	36.5
Healthy	36.7
Healthy	36.8
Sick	38.0
Sick	38.2
Sick	38.4

Naive Bayes – numerical cont'd

Prior x **Likelihood**

$$P(C_1) = \frac{3}{6} = 0.5$$

$$P(C_2) = \frac{3}{6} = 0.5$$

$$P(x|C_k) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(x-\mu_k)^2}{2\sigma_k^2}\right)$$

Naive Bayes – numerical cont'd

Healthy (C_1)

$$\mu_1 = \frac{36.5 + 36.7 + 36.8}{3} = 36.67$$

$$\sigma_1^2 = \frac{(36.5 - 36.67)^2 + (36.7 - 36.67)^2 + (36.8 - 36.67)^2}{3}$$

$$\sigma_1^2 \approx 0.0156$$

Sick (C_2)

$$\mu_2 = \frac{38.0 + 38.2 + 38.4}{3} = 38.2$$

$$\sigma_2^2 = \frac{(38.0 - 38.2)^2 + (38.2 - 38.2)^2 + (38.4 - 38.2)^2}{3}$$

$$\sigma_2^2 \approx 0.0267$$

Naive Bayes – numerical cont'd

Classify 37.5:

Likelihood for Healthy

$$P(x|C_1) = \frac{1}{\sqrt{2\pi(0.0156)}} \exp\left(-\frac{(37.5 - 36.67)^2}{2(0.0156)}\right)$$

$$P(x|C_1) \approx 0.00003$$

Likelihood for Sick

$$P(x|C_2) = \frac{1}{\sqrt{2\pi(0.0267)}} \exp\left(-\frac{(37.5 - 38.2)^2}{2(0.0267)}\right)$$

$$P(x|C_2) \approx 0.11$$

$$P(C_1|x) \propto P(x|C_1)P(C_1) = 0.00003 \times 0.5$$

$$P(C_2|x) \propto P(x|C_2)P(C_2) = 0.11 \times 0.5$$